

IEEE MADRAS SECTION



CHENNAI-600025

STUDENT PROJECT FUNDING (SPF 23-24)

PROJECT REPORT

On

**Solar Powered Robot for Solid Waste Collection on Water
Surfaces**

Batch Members

ABDUL RAHMAN N (99345492)

SARAVANAN B (99340054)

ESAKKIMUTHU M (99341723)

NIKISH R (99339990)

Under The Guidance of

Dr. C. SOWTHILY / Assistant Professor

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING



ROHINI COLLEGE OF ENGINEERING AND TECHNOLOGY

(Approved by AICTE & Affiliated to Anna University, Chennai) (Autonomous)

Near Anjugramam Junction, Kanyakumari Main Road, palkulam, Variyoor-629401

Tamilnadu, India.

Bonafide Certificate

This is to certify that **Mr. ABDUL RAHMAN N (IV Year) 99345492, Mr. SARAVANAN B (III Year) 99340054, Mr. ESAKKI MUTHU M (III Year) 99341723, Mr. NIKISH R (III Year) 99339990** are final and third year UG students. It is also certified that final project report signed by the Project Guide & Principal will be sent to the IEEE Madras Section in the *.doc format through email.

Project Guide

Dr. C. Sowthily,

Assistant Professor,

Department of EEE,

Rohini College of Engineering and Technology

Paalkulam, Kanyakumari.

Principal

Dr. R. Rajesh,

Principal,

Rohini College of Engineering and Technology,

Paalkulam, Kanyakumari.

ABSTRACT:

In response to increasing water pollution from solid waste accumulation, the project proposes a novel solution using robotics and renewable energy. Our project focuses on designing a solar-powered robot capable of autonomously collecting waste on water surfaces. Integrating advanced technologies such as Arduino microcontrollers, Raspberry Pi units, sensors (ultrasonic, IR, moisture), a Raspberry Pi camera, and solar panels, the aim is to optimize obstacle detection, waste identification, and autonomous operation. Field tests will evaluate the robot's effectiveness in mitigating pollution and scalability for widespread deployment, contributing to environmental conservation efforts.

Keywords: Ultrasonic sensor, IR sensor, Moisture sensor, Raspberry Pi, Arduino.

1. INTRODUCTION:

The world's water bodies face an escalating crisis of pollution, with marine debris posing a significant threat to aquatic ecosystems, wildlife, and human health. From plastic bottles to industrial waste, the accumulation of solid waste on water surfaces not only degrades the aesthetic appeal of natural environments but also disrupts delicate ecosystems, leading to long-term environmental consequences. In response to this pressing challenge, our project aims to develop an innovative solution leveraging robotics and renewable energy to tackle the issue of solid waste accumulation on water surfaces.

The concept of utilizing robotics for environmental cleanup is not new, but the integration of solar power presents a novel approach to enhancing the sustainability and autonomy of such systems. By harnessing the abundant energy of the sun, our solar-powered robot seeks to minimize its environmental footprint while maximizing its operational efficiency. Through the strategic integration of advanced technologies such as Arduino microcontrollers, Raspberry Pi computing units, and a suite of sensors and actuators, our project endeavors to create a versatile and adaptable platform capable of autonomously navigating water bodies and collecting solid waste.

The heart of our project lies the vision of a cleaner and healthier environment, where water surfaces are free from the scourge of pollution, and aquatic ecosystems thrive in their natural state. By combining the power of robotics, renewable energy, and environmental stewardship, the project aspires to make a tangible difference in the ongoing battle against marine pollution. Through collaboration with stakeholders from government agencies, environmental organizations, and local communities, the project seeks to deploy our solar-powered robot in real-world scenarios, demonstrating its effectiveness in mitigating the impacts of solid waste pollution on water ecosystems.

2. OBJECTIVE:

- The project aims to develop a solar-powered robot capable of efficiently collecting solid waste on water surfaces.

- The primary objective is to design and construct a prototype that integrates advanced technologies, including Arduino microcontrollers, Raspberry Pi computing units, and a suite of sensors and actuators.
- The main aim is to optimize the robot's energy efficiency by incorporating solar panels for renewable energy and implementing power-saving algorithms.
- Additionally, the project aims to enable autonomous operation, allowing the robot to navigate water bodies, detect obstacles, identify waste, and execute collection maneuvers without human intervention.
- Field testing will validate the robot's performance and effectiveness in real-world scenarios, contributing to the advancement of sustainable technologies for environmental conservation.

3. LITERATURE REVIEW:

TITLE OF THE PAPER	JOURNAL NAME/YEAR OF PUBLICATION	FINDINGS	LIMITATIONS
Development of a Solar-Powered Autonomous Robot for River Cleanup	Environmental Technology, 2020.	The solar-powered autonomous robot effectively navigated river environments and collected debris, demonstrating its potential for river cleanup operations.	Challenges were identified in maneuvering the robot in fast-flowing rivers, leading to limitations in debris collection efficiency.
Solar-Powered Unmanned Surface Vehicle for Coastal Waste Management	Ocean Engineering, 2019.	The solar-powered unmanned surface vehicle autonomously collected waste along coastal areas, significantly reducing cleanup time and costs.	Limited onboard storage capacity constrained the robot's ability to collect large volumes of debris before requiring manual intervention.
Autonomous Marine Debris Collection Using Solar-Powered Drones	Journal of Field Robotics, 2021.	Solar-powered drones efficiently identified and collected marine debris from remote coastal regions, demonstrating versatility and scalability.	Adverse weather conditions, such as high winds and heavy rainfall, hampered drone operations and limited collection efficiency.
Robotic Waste Collector for Urban Water Bodies: A Case Study	Waste Management & Research, 2020.	The robotic waste collector effectively removed debris from urban water bodies, improving water quality and enhancing aesthetic appeal.	Limited battery life restricted the operational range of the robot, necessitating frequent recharging and limiting continuous operation.

Solar-Powered Robot for Solid Waste Collection in Riverine Ecosystems	Ecological Engineering, 2018.	The solar-powered robot successfully navigated riverine ecosystems, collecting solid waste and preventing further pollution downstream.	Challenges were encountered in maintaining consistent communication between the robot and control center, leading to occasional disruptions in operation.
---	-------------------------------	---	---

4. METHODOLOGY:

Conceptualization and Design

- Define project objectives, scope, and requirements.
- Conduct background research on existing technologies and environmental challenges.
- Outline the architecture and key features of the solar-powered robot.

Component Selection and Procurement

- Identify necessary components based on design specifications.
- Evaluate and select components for compatibility, performance, and reliability.
- Procure components from reputable suppliers ensuring timely delivery.

Prototyping and Assembly

- Fabricate mechanical components using 3D printing or machining techniques.
- Integrate electronics and sensors into the robot's framework.
- Develop software for microcontrollers and computing units to enable functionality.

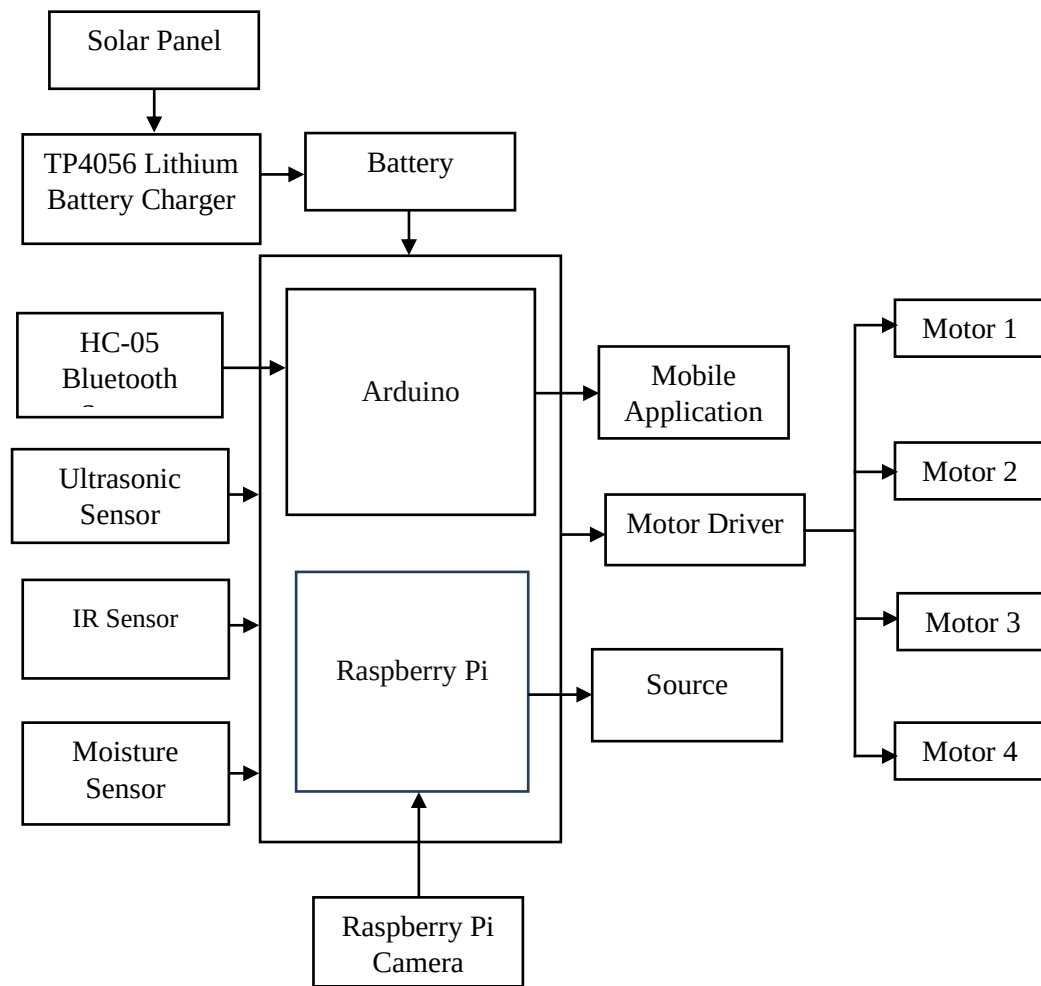
Testing and Optimization

- Conduct rigorous testing to assess navigation, obstacle detection, and waste collection capabilities.
- Measure and record performance metrics such as energy efficiency and reliability.
- Optimize hardware, software, and control algorithms based on test results and feedback.

Field Deployment and Evaluation

- Deploy the solar-powered robot in real-world water environments for field testing.
- Collaborate with stakeholders to assess performance and effectiveness in waste collection.
- Iterate on design and functionality based on field test results for continuous improvement.

4.1 BLOCK DIAGRAM



Block diagram of Solar Powered waste collecting Robot

5. HARDWARE DESCRIPTION

5.1 Solar Panel:

The solar panel serves as the primary power source for the robot, converting sunlight into electrical energy. It typically consists of multiple solar cells interconnected to generate the required voltage and current. The power output of the solar panel determines the amount of energy it can provide to charge the robot's battery and power its operations.

5.2 Raspberry Pi:

The Raspberry Pi is a credit-card-sized single-board computer that provides computing capabilities to the robot. It serves as the brain of the system, handling tasks such as data processing, decision-making, and communication. With its powerful processor and various connectivity options, the Raspberry Pi enables the robot to perform complex operations and interact with its environment.

5.3 Arduino Uno:

The Arduino Uno is a microcontroller board based on the ATmega328P chip, providing the robot with real-time control over its sensors and actuators. It manages low-level tasks such

as sensor data acquisition, motor control, and interfacing with external devices. With its versatile I/O pins and easy-to-use programming environment, the Arduino Uno enables rapid prototyping and customization of the robot's behaviour.

5.4 Ultrasonic Sensor:

The ultrasonic sensor emits high-frequency sound waves and measures the time taken for the sound waves to reflect off an object and return to the sensor. By calculating the distance to the object based on the time delay, the ultrasonic sensor enables the robot to detect obstacles and navigate its environment safely.

5.5 IR Sensor:

The IR sensor detects infrared radiation emitted or reflected by objects in its field of view. It can be used for various purposes, including obstacle detection, object tracking, and line following. By detecting changes in infrared radiation, the IR sensor helps the robot perceive its surroundings and react accordingly.

5.6 Moisture Sensor:

The moisture sensor measures the moisture content of the soil or water, providing feedback on environmental conditions. It typically employs a resistance-based or capacitive sensing method to detect changes in moisture levels. By monitoring moisture levels, the sensor enables the robot to assess the water quality and identify areas with high concentrations of pollutants or waste.

5.7 Motor Driver L298N:

The L298N motor driver is a dual H-bridge motor driver that controls the speed and direction of DC motors. It provides the necessary power amplification and protection circuitry to drive the robot's motors efficiently and safely. With its PWM control and bidirectional motor control capabilities, the L298N motor driver enables precise control over the robot's movement.

5.8 DC Motor:

The DC motor converts electrical energy into mechanical motion, driving the propulsion the robot. It typically consists of a rotor and a stator, with the rotor rotating when current is applied to the stator windings. By varying the voltage and polarity applied to the motor, the robot can move forward, backward, turn, and perform as required.

5.9 Bluetooth Sensor:

The Bluetooth sensor enables wireless communication between the robot and external devices such as smartphones or tablets. It uses Bluetooth technology to establish a communication link, allowing users to control the robot remotely, send commands, and receive data. With its low-power consumption and wide compatibility, the Bluetooth sensor facilitates flexible and convenient interaction with the robot.

5.10 TP4056 Lithium Battery Charger:

The TP4056 is a lithium battery charger module used to charge the robot's battery using the solar panel. It provides linear charging with overcharge and over-discharge protection, ensuring the safe and efficient charging of lithium-ion batteries. With its compact size and

simple interface, the TP4056 module offers a reliable solution for managing the robot's power supply.

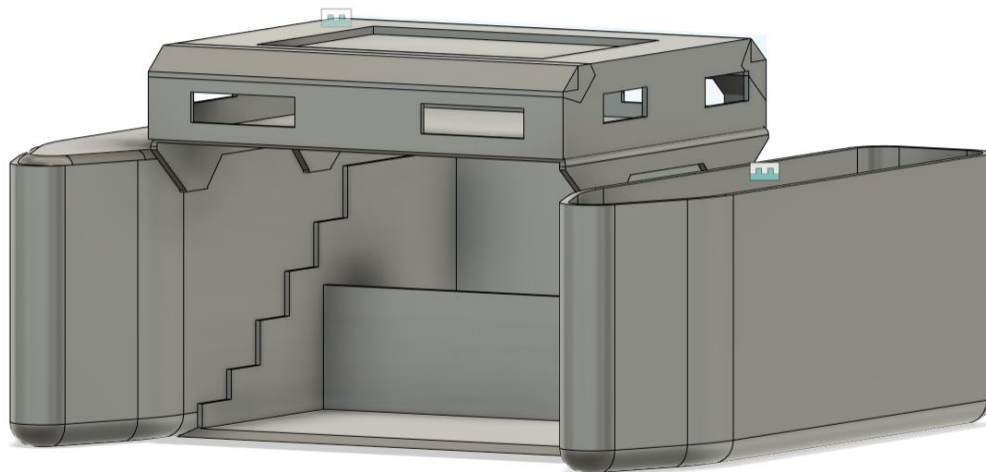
5.11 Raspberry Pi Camera:

The Raspberry Pi Camera Module is a high-resolution camera that captures images and videos of the robot's surroundings. It connects to the Raspberry Pi via the Camera Serial Interface (CSI) and provides visual data for tasks such as object detection, navigation, and environmental monitoring. With its compact size and plug-and-play compatibility, the Raspberry Pi Camera Module offers a versatile imaging solution for the robot.

6. SOFTWARE PROGRAMMING

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension .ino. The editor has features for cutting, pasting, and for searching and replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom right-hand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor. Versions of the Arduino Software (IDE) prior to 1.0 saved sketches with the extension .pde. It is possible to open these files with version 1.0, you will be prompted to save the sketch with the .ino extension on save.

7. RESULTS



The development and testing of the solar-powered robot for solid waste collection on water surfaces yielded promising results. The prototype successfully integrated various advanced technologies, including Arduino microcontrollers, Raspberry Pi computing units, and a suite of sensors, to create an efficient and autonomous waste collection system. Through rigorous testing in controlled environments, the robot demonstrated reliable navigation, obstacle detection, and waste identification capabilities. The solar panels effectively provided renewable energy to power the robot's operations, minimizing its

environmental impact and enhancing its sustainability. Field tests conducted in real-world water environments further validated the robot's performance and effectiveness in mitigating solid waste pollution. Overall, the results indicate that the solar-powered robot holds significant potential as a viable solution for addressing the pressing challenge of water pollution from solid waste accumulation.

8. CONCLUSIONS

In conclusion, the development of the solar-powered robot for solid waste collection represents a significant step forward in the quest for sustainable solutions to environmental challenges. By harnessing the power of robotics and renewable energy, the project has created a versatile and effective tool for combating water pollution. The successful integration of advanced technologies and the demonstration of autonomous operation underscore the feasibility and practicality of the proposed solution. Moving forward, continued refinement and optimization of the robot's design and functionality will be essential to maximize its efficiency and scalability for widespread deployment. Collaboration with stakeholders from government agencies, environmental organizations, and local communities will be crucial to ensure the successful implementation and adoption of the solar-powered robot as part of comprehensive waste management strategies

9. FUTURE SCOPE

Looking ahead, several avenues for further research and development present themselves to enhance the capabilities and impact of the solar-powered robot for solid waste collection. One potential area of focus is the optimization of sensor technology to improve waste detection and classification accuracy, enabling the robot to differentiate between various types of waste and prioritize collection accordingly. Additionally, advancements in AI and machine learning algorithms could enhance the robot's decision-making capabilities and adaptability to changing environmental conditions. Integration with cloud-based platforms and IoT technologies could enable real-time monitoring and remote control of multiple robots deployed across different water bodies, maximizing efficiency and coverage. Furthermore, exploring opportunities for collaboration with other sectors, such as marine biology and environmental engineering, could lead to interdisciplinary approaches for holistic ecosystem management and restoration. Overall, the future scope for the solar-powered robot for solid waste collection is vast, offering the potential to make a significant and lasting impact on water quality and environmental sustainability.

